CSE - 641 Computer Vision

Weekly Project Report – 1

Ahmedabad University

Genotype-to-Phenotype Modeling of Alzheimer’s Disease Using MRI Radiomics and Polygenic Risk Scores

Professor: Dr. Mehul Raval

Teaching Assistant: Shalvi Modi

Team Members

Name	Enrollment Number
Yash Shah	AU2340078
Ura Modi	AU2340031
Dhriti Gandhi	AU2340030
Devashree Parikh	AU2340067

***Abstract*—Alzheimer disease is a disorder that affects the brain leading to loss of memory and progressive impairment of the thinking ability. Studies indicate that genetic factors as well as changes in the brain structure are significant in the pathogenesis of Alzheimer disease. However, the correlation between the inheritable genetic risk and the change in the brain structure is not well understood.**

The presented project seeks to examine the association between genetic vulnerability and brain structure by means of Polygenic Risk Scores (PRS) and MRI-based radiomic characteristics. The features of important brain areas including the hippocampus and the entorhinal cortex, will be extracted using MRI scans of the Alzheimer Disease Neuroimaging Initiative (ADNI) dataset. To determine inherited genetic risk, PRS will be calculated. Machine learning models will be then used to identify the relationship between genetic risk and features of MRI.

The aim of the study is to gain more insight into the role of genetic factors in the early brain changes of the Alzheimer disease potentially before symptoms appear.

I. Introduction

Alzheimer disease is an advanced disorder of the brain which gradually impairs the memory, ability to think and general brain performance. It is connected with when brain tissue is gradually lost throughout time. These changes are usually studied by structural magnetic resonance imaging (MRI), particularly in those parts of the brain that become affected at an early stage in the disease, the hippocampus, entorhinal cortex and temporal lobes [1]. Meanwhile, genome wide association studies of large sizes have enhanced the insights into the genetic etiology of Alzheimer. The genetic risk variants that have been identified through such studies are too many and predispose an individual with the probable cause of the disease [2]. These variants may be used together in a polygenic risk score (PRS) i.e., to sum up inherited genetic risk in a single value [3]. Both MRI markers and PRS can be used to understand the disease, but this is in two dimensions- brain structure and genetic risk.

Recently, scientists have begun using genetic and imaging data together in order to have greater insight into the influence of inherited risk on the brain prior to the emergence of symptoms. The goal of imaging genetics research is to investigate genetic risk or structure brain interactions during the early phases of the disease [4]. Moreover, radiomics enables the derivation of hundreds of shape and texture features of an MRI scan that are no longer measured in terms of volume [5]. With the convergence of PRS and radiomic features using machine learning models based on MRI, it is possible to investigate the effects of genetic risk on the brain structure. This will be a way of gaining a more clear evidence on how genetic factors lead to early structural changes in Alzheimer disease.

II. Dataset Description

The data used in this project is obtained from the **Alzheimer's Disease Neuroimaging Initiative (ADNI)**, a multi-centered longitudinal large-scale study established to identify and validate biomarkers for the early detection and monitoring of Alzheimer's disease progression.

The study participants are categorized into three primary groups:

- Cognitively Normal (CN)
- Mild Cognitive Impairment (MCI)
- Alzheimer's Disease (AD)

ADNI provides multimodal data, including:

- Structural Magnetic Resonance Imaging (MRI)
- Positron Emission Tomography (PET) scans (FDG-PET and Amyloid PET)
- Cerebrospinal Fluid (CSF) biomarker values
- Genetic data (e.g., APOE genotype)
- Neuropsychological assessment scores (e.g., MMSE, CDR, ADAS-Cog)
- Demographic information

The data were collected across multiple phases, namely:

- ADNI-1
- ADNI-GO
- ADNI-2
- ADNI-3

These phases enable longitudinal analysis of disease progression over time. The dataset is cross-standardized across acquisition sites, ensuring consistency and reliability.

Due to its comprehensive and longitudinal nature, the ADNI dataset is widely used in machine learning and deep learning research for classification, progression prediction, and biomarker discovery in Alzheimer's disease studies.

III. Literature Review

A. *Longitudinal structural MRI-based deep learning and radiomics features for predicting Alzheimer's disease progression*

The authors in this paper show how a longitudinal MRI-based paradigm can be utiliized to forecast the subjects who will progress to Alzheimer disease from mild cognitive impairment (MCI). They build a two-stage deep learning model on the T1-weighted images of the Alzheimer Disease Neuroimaging Initiative dataset. They first train a 3D ResNet18 model on individual visit MRI volumes to enable it to learn spatial brain features related to disease risk with a survival-based ranking loss. The learned feature representations are then used as input to a time-varying LSTM that trains to learn multiple MRI scans over time, accounting for irregular time intervals between visits. The attention mechanism identifies the most relevant time points. They also discard gray matter radiomics features like shape, intensity, and texture features, and discard them using PCA, and then combine them with deep features. The final model is being placed within a survival analysis framework to estimate individualized risk of time-to-conversion, rather than just binary classification.

B. Predicting Conversion from Mild Cognitive Impairment to Alzheimer's Disease Using a Vision Transformer and Hippocampal MRI Slices

The deep learning models are employed to predict the development of Mild Cognitive Impairment (MCI) to Alzheimer's Disease (AD) from MRI scans. Most of the existing studies were conducted using Convolutional Neural Networks (CNNs) because they possess the capability to learn spatial information from images of the brain. Some studies have been demonstrated to possess high accuracy, but there were problems such as data leakage, where data from the same patient was used for training and testing, which could potentially lead to biased results.

Recently, Vision Transformers (ViTs) have been proposed for image processing. Although Vision Transformers have been demonstrated to possess high accuracy for Alzheimer's classification, few studies have been conducted to explore their application for predicting MCI to AD conversion.

This paper continues from the existing studies by employing a Vision Transformer model that was pretrained on hippocampal MRI slices. It strictly separates data for each subject and compares the results to a 3D CNN model. The results indicate that Vision Transformers possess the capability to produce similar results to CNN models.

IV. Work Completed in Week-1

In Week 1, our group studied the basics of Alzheimer's Disease (AD) and understood the three clinical categories: Cognitively Normal (CN), Mild Cognitive Impairment (MCI), and Alzheimer's Disease (AD).

- We learned that the disease typically progresses in the sequence: CN $\rightarrow$ MCI $\rightarrow$ AD.
- We also studied the main biological changes associated with the disease, including amyloid-beta plaques, tau protein tangles, and hippocampal brain atrophy.
- Additionally, we explored the Alzheimer's Disease Neuroimaging Initiative (ADNI) dataset and understood that it is a multi-center, longitudinal study designed for early detection and monitoring of Alzheimer's Disease progression.

All in all, the week was spent on developing the clear vision of the dataset and the issue prior to beginning to implement the models.

V. Plan for Week-2

During Week-2, we will further refreeze our grasp of current research and will start putting it into practice. The tasks to be done are as follows:

- In Week 2, we will identify our problem, for instance, patient classification based on CN, MCI, and AD. We will then identify the type of data that we will use, for instance, MRI, PET, or multi-modal data.
- We will explore the ADNI dataset.
- Explore additional enhancements with the help of literature.

References

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